

WEEK 10

# Single cells and biological replication

Do more cells solve a shortage of donors?

## By the end, you should be able to...

Explain a cell-by-gene matrix, a barcode and a UMI.

Separate exploratory markers from condition-level inference.

Aggregate counts by donor and cell type and identify the independent sample size.

# Reconnect with the biology

Bulk RNA-seq averages mixtures. Single-cell measurements can separate cell populations, but cells remain nested within samples.

**Think silently -> explain to a neighbour -> check the measurement.**

# From a tissue to many cell profiles

**01**

Cells or nuclei are captured and their RNA-derived molecules labeled.

>

**02**

Cell barcodes identify capture compartments.

>

**03**

UMIs help distinguish original molecules from amplification copies, subject to error and correction.

# The matrix has an orientation

| Object      | Meaning                               |
|-------------|---------------------------------------|
| obs         | Cell / observation metadata           |
| var         | Gene / variable metadata              |
| X or layers | A documented measurement or transform |

AnnData convention: observations are rows; variables are columns.

For a typical single-cell dataset: cells x genes.

Check what X and layers contain; never assume every matrix stores raw counts.

# A zero is not proof of absence

A molecule may not be captured or sequenced.

Low counts can reflect biology, sampling or quality.

Avoid claiming a gene is absent from a tissue based on a few zero entries.

# Quality control is contextual

1

Inspect counts, detected genes, mitochondrial fraction and sample patterns.

2

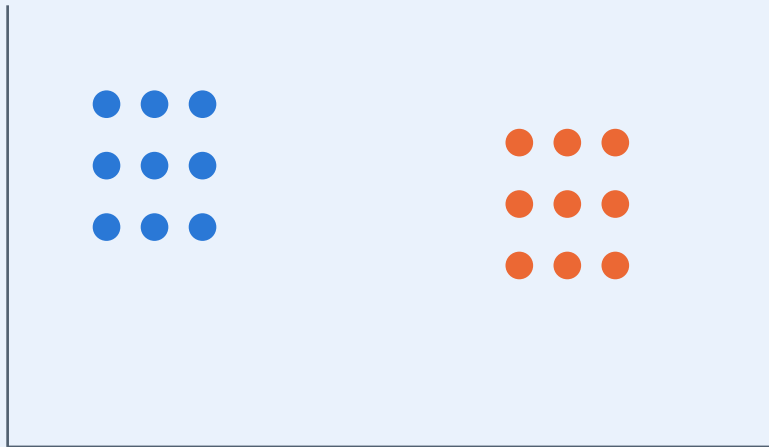
A universal mitochondrial cutoff is not appropriate for all tissues and protocols.

3

Doublets and ambient RNA can distort cell profiles.

# Clustering is a hypothesis generator

## Illustrative 2D projection



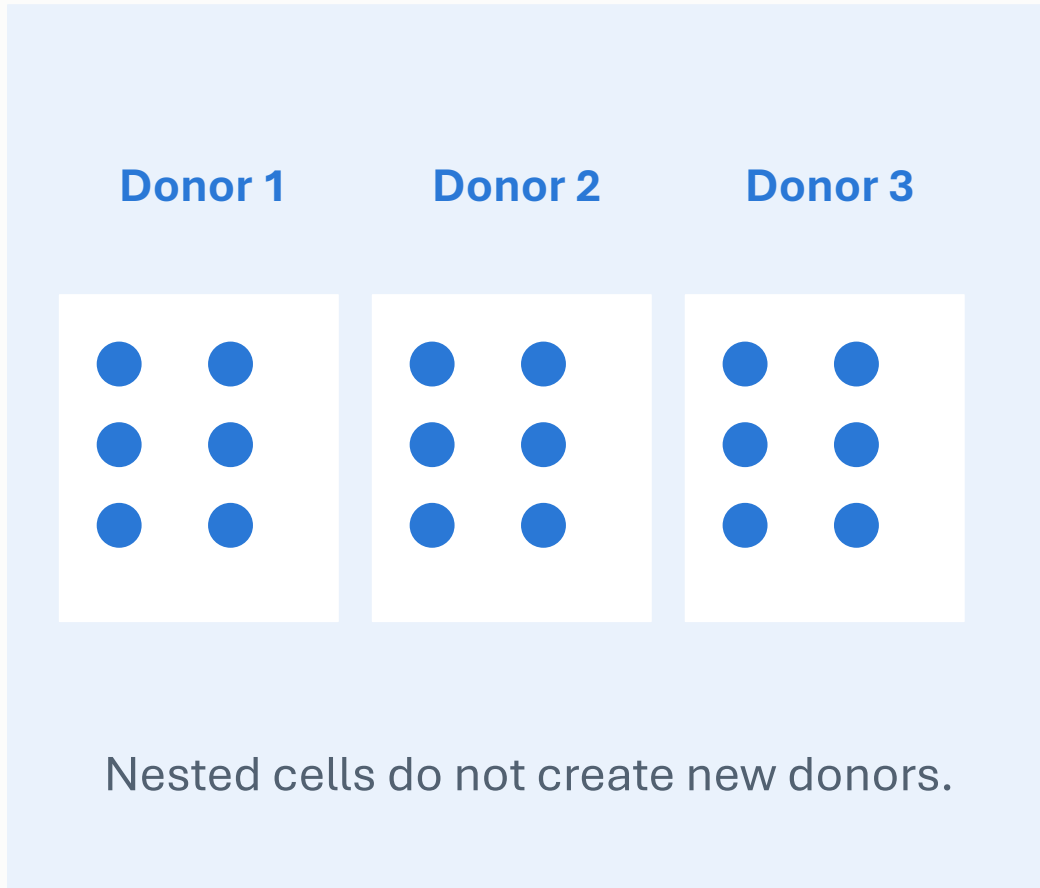
A displayed group is a hypothesis, not a diagnosis.

A cluster depends on the representation and chosen resolution.

Use multiple markers and sample context to propose a cell identity.

A cluster label and a statistical significance claim are different things.

# For condition comparisons, keep donor identity



Many cells from one donor share biological and technical factors.

Treating them as independent donors can underestimate uncertainty.

Between-donor replication is essential for a donor-level comparison.

# Pseudobulk retains the biological unit

01

Within each donor and cell type, sum raw counts.

>

02

The resulting columns or rows represent donor-cell-type samples.

>

03

A count model can then represent the experimental design, including pairing when appropriate.

# Pseudobulk is not a magic rescue

One donor per condition still lacks between-donor replication.

Very few donors imply limited precision and power.

A donor-aware mixed model is another possible approach; method choice depends on the question.

## Worked example: aggregate, do not multiply donors

A cell type has counts [2,3] in donor A and [4,1,2] in donor B.

**Predict the result and one limitation before calculating.**

# Work through the reasoning

Pseudobulk count for A = 5.

Pseudobulk count for B = 7.

There are two donor-cell-type samples, not five independent donors.

## Pause: make a defensible claim

A study measures 10,000 cells from one control donor and one treated donor. Would doubling the cells enable a defensible population-level treatment comparison?

**First answer alone. Then compare the evidence, not just the conclusion.**

# A defensible response

No. It improves within-donor sampling but does not add independent donors or estimate general between-donor variability.

# Today's practical

W10: aggregate a synthetic cell table, compare cell and donor sample sizes, and explain why a cell-level shortcut fails. The lab uses no AnnData installation.

| Minutes | Action                |
|---------|-----------------------|
| 0-10    | Predict + set up      |
| 10-30   | Run + inspect + check |
| 30-40   | Explain + compare     |
| 40-45   | Save + submit / exit  |

# Five terms to use precisely

- 1 barcode
- 2 UMI
- 3 cell type
- 4 pseudobulk
- 5 donor

# Exit ticket

What did we actually measure or calculate?

What is one assumption or limitation of that result?

What check would you perform before sharing it?

# Go further only after the core is clear

Optional: differential abundance and paired donor designs; no atlas integration pipeline is required.

**Not required for the core grade. No advanced model training is expected.**

# Read the evidence, not only the slide

S09 Love, Huber and Anders (2014). Moderated estimation of fold change and dispersion for RNA-seq data with DESeq2.

S12 Squair et al. (2021). Confronting false discoveries in single-cell differential expression.

S13 AnnData documentation: annotated data matrices.

S27 EMBL-EBI: Single cell RNA-seq analysis using Python materials (2026).